

QPOS iOS SDK

Integration Guide

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Document Overview

This document is the official integration guide for the QPOS iOS SDK.

It provides comprehensive, step-by-step instructions to assist developers in integrating QPOS payment terminal devices—including QPOS, QPOS Reader — into **iOS applications**. The guide covers key areas such as system prerequisites, SDK API usage, transaction workflows, and general operational guidelines. Its objective is to enable developers to complete the integration efficiently while ensuring that payment functionality remains secure, reliable, and stable.

1. Version History

Version	Date	Author	Main Modifications
1.0	2013.07.12	Longhui.xiao	Initial added
1.1	2013.12.05	Junan Zhang	
2.0	2014.01.05	Longhui.xiao	
2.1	2014.02.12	Wang Xu	Minor bug fix
2.2	2014.03.11	Longhui Xiao	Add ICC APDU
2.3	2014.03.26	Longhui Xiao	Add set the sleep time interface
2.4	2014.04.08	Longhui Xiao	Add update work key interface
2.5	2015.02.13	Qingfu Zeng	Perfect the documents
2.6	2015.03.16	Longhui Xiao	Review modifies the document
2.7	2015.09.17	Longhui Xiao	Added resetPosStatus interface. Perfect the documents
2.8	2022.08.23	Ziwei Wang	Add an Example
2.9	2025.11.06	Shasha Lv	Adjust file format and structure

2. Preface

2.1 Purpose & Scope

Purpose: This document standardizes the integration process for the QPOS iOS SDK.

It clarifies the usage of SDK APIs, transaction workflows, and troubleshooting methods, ensuring that developers can correctly integrate QPOS devices with iOS applications to achieve secure and stable payment transactions.

Scope: This document covers the QPOS series devices (QPOS, QPOS Reader) and their corresponding iOS SDK. It includes system environment requirements, SDK framework structure, permission configurations, API parameter descriptions, transaction flow diagrams, and delegate callback methods. It does **not** cover hardware maintenance or servicing of QPOS devices.

2.2 Intended Audience

- iOS application developers responsible for integrating mobile payment functions.
- Technical support engineers who provide QPOS SDK integration guidance.
- Testers who verify the compatibility and stability of QPOS device connections.

3. Glossary & Definitions

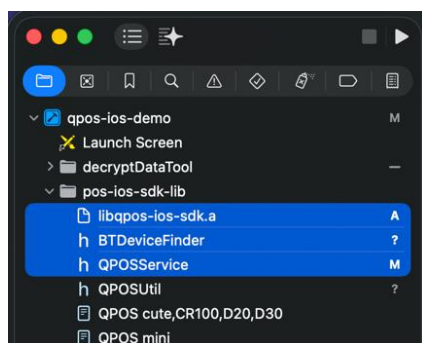
Abbreviation	Full Name / Definition
SDK	Software Development Kit (a toolset for integrating QPOS devices with applications)
QPOS	Mobile payment card reader with PIN pad (supports multiple communication modes)
QPOS Reader	Mobile payment card reader without PIN pad (supports Audio/USB communication)
EMV	Europay, MasterCard, Visa (a global standard for chip card payments)
ICC	Integrated Circuit Card (commonly known as "chip card")
APDU	Application Protocol Data Unit (data interaction protocol between card and terminal)
DUKPT	Derived Unique Key Per Transaction (a key management standard for secure transactions)
KSN	Key Serial Number (unique identifier for key derivation)
IPEK	Initial PIN Encryption Key (initial key for PIN encryption)
TLV	Tag-Length-Value (a data encoding format used in EMV transactions)

4. SDK Overview

This article is the integration development guide document for the Dspread unified SDK standard. It is intended to guide users on how to use the SDK, assuming that the reader is already familiar with the basic usage of the IDE(Xcode) and has a certain foundation in iOS programming.

4.1 Implement SDK Method

Please add “libqpos-ios-sdk.a”、“BTDeviceFinder.h”、“QPOSService.h” these three files in your project.



4.2 Supported System Versions & IDE

- The SDK supports a minimum deployment version of iOS 9.0
- The SDK only supports integration with Xcode.

4.3 iOS permission

```
<key>NSBluetoothAlwaysUsageDescription</key>
<string>This app requires Bluetooth access to connect to POS devices.</string>

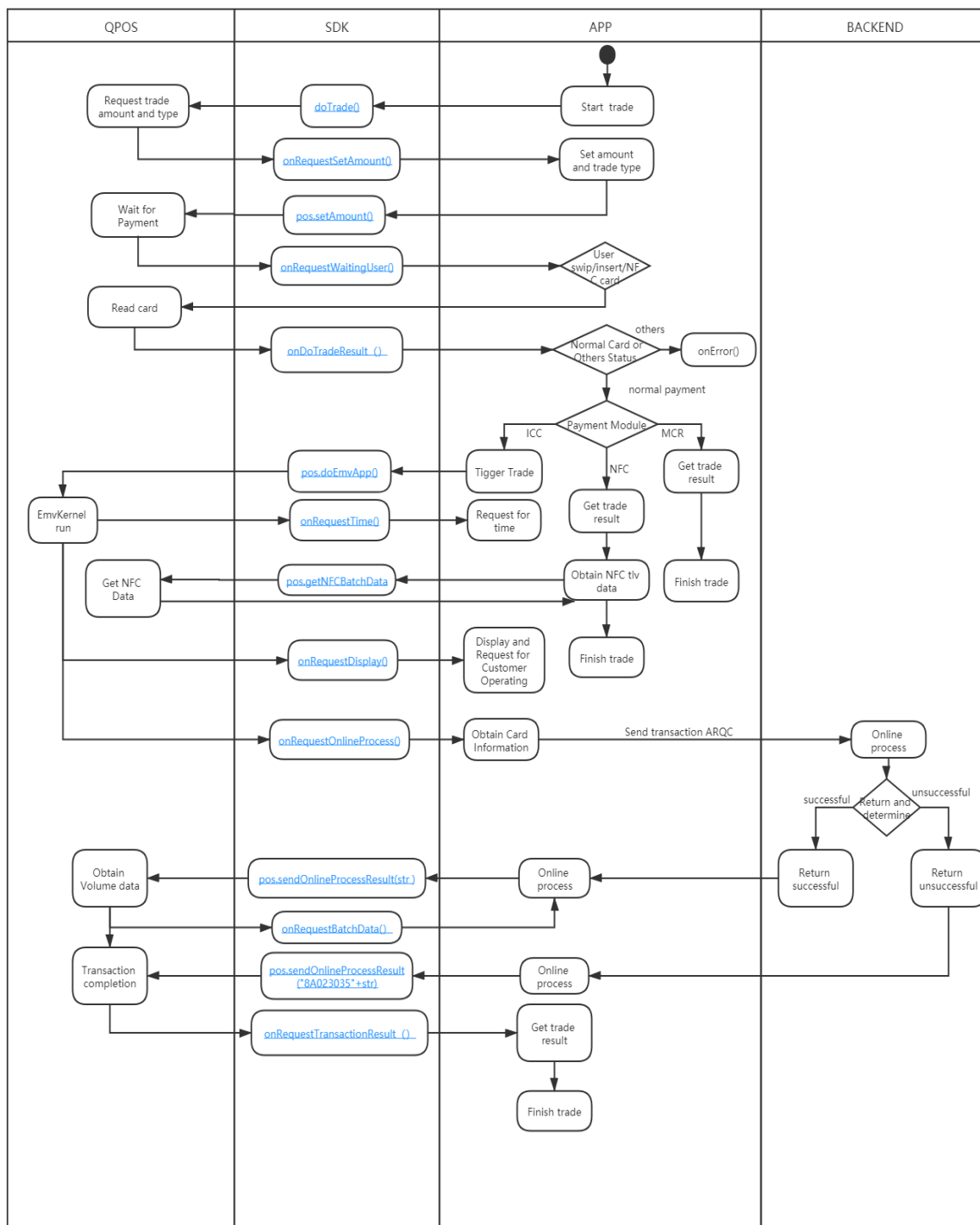
<key>NSMicrophoneUsageDescription</key>
<string>The microphone is required for audio-based communication with the POS
terminal.</string>
```

The QPOS SDK requires specific permissions to access Bluetooth and Audio (for Bluetooth scanning). Add the following permissions to the **Info.plist** file in your application:

4.4 API Reference

This section introduces the core APIs offered by the QPOS SDK, organized by functionality.

Before reviewing the API specifications, please refer to the transaction flowchart below to gain a clearer understanding of the overall process for application integration.



4.4.1 Connection APIs

These APIs are used to initialize, connect, and disconnect QPOS devices, supporting Audio, Bluetooth communication modes.

● Init QPOSService

API Methods	Description
+(QPOSService *)sharedInstance;	Get the QPOSService instance
- (void)setDelegate:(id<QPOSServiceListener>)aDelegate;	QPOSServiceListener is a listener/delegate interface in the QPOS iOS SDK used to receive POS device status updates, transaction process callbacks, and system event notifications.

● Bluetooth connection

API Methods	Description
-(void)scanQPos2Mode:(NSInteger)timeout;	Scans for dual-mode Bluetooth devices
-(void)stopQPos2Mode;	Stops scanning for dual-mode Bluetooth devices
-(BOOL)connectBT:(NSString*)bluetoothName;	Connects using Classic Bluetooth
-(void)disconnectBT;	Disconnects the current device

Audio connection

API Methods	Description
-(void)startAudio;	Start the audio instance for playing, recording and modulating
-(void)stopAudio;	Stop the audio instance for playing, recording and modulating

sharedInstance

Item	Details
Signature	+(QPOSService *)sharedInstance;

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Item	Details
Inputs	None
Outputs	Return the QPOSService instance
Example	[QPOSService sharedInstance]

setDelegate

Item	Details
Signature	-(void)setDelegate:(id<QPOSServiceListener>)aDelegate;
Inputs	listener: an instance of QPOSServiceListener
Outputs	None
Example	[pos setDelegate:self];

scanQPos2Mode

Item	Details
Signature	-(void)scanQPos2Mode:(NSInteger)timeout;
Inputs	The time out for scanning device
Outputs	None
Example	[pos scanQPos2Mode:30]
Callback	onBluetoothName2Mode

stopQPos2Mode

Item	Details
Signature	-(void)stopQPos2Mode;
Inputs	None
Outputs	None
Callback	None

connectBT

Item	Details
Signature	-(BOOL) connectBT: (NSString *)bluetoothName;
Inputs	Bluetooth name.
Outputs	Conenction result
Example	[pos connectBT:@"MPOS0010900318"];
Callback	onRequestQposConnected

disconnectBT

Item	Details
Signature	-(void)disconnectBT()
Inputs	None
Outputs	None
Callback	onRequestQposDisconnected

startAudio

Item	Details
Signature	-(void)startAudio;
Inputs	None
Outputs	None
Example	[pos startAudio]
Callback	onRequestQposConnected

stopAudio

Item	Details
Signature	-(void)stopAudio;

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Item	Details
Inputs	None
Outputs	None
Example	[pos stopAudio]
Callback	onRequestQposDisconnected

4.4.2 Transaction APIs

Provides the relevant APIs for executing transactions and details the associated transaction processes. Below is a comprehensive list of these APIs:

API Methods	Description
-(void)setCardTradeMode:(CardTradeMode)aCardTMode;	Set the card reading mode allowed by the device
-(void)doTrade(int timeout)	Send the command as swiping/inserting/tapping card to POS.
-(void)setAmountPoint:(BOOL)amoutPoint;	Set the amount and transaction type required for EMV transaction
-(void)cancelSetAmount()	Cancel the process about setting amount
-(void)sendTime: (NSString *)aterminalTime;	Send current time.
-(void)doEmvApp: (EmvOption)aemvOption;	Send the command as executing the EMV transaction flow to POS
-(void)selectEmvApp: (NSInteger)index;	Select one application of the application list returned from EMV kernel, then set the application ID to EMV kernel
-(void)cancelSelectEmvApp()	Cancel the process about setting the application
-(void)sendPinEntryResult:(NSString *)pin;	Send Pin Entry Result to terminal

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API Methods	Description
-(void)sendCvmPin:(NSString *)pin isEncrypted:(BOOL)isEncrypted;	If input is True will send encrypt pinblock (ISO-4) to terminal and if not will send plaintext PIN.
-(void)cancelPinEntry;	Cancel Pin Entry
-(void)sendOnlineProcessResult(String tlv)	Send the tlv data that returned by the issuer to terminal.
-(void)getNFCBatchData:(void(^)(NSDictionary *dict))nfcBatchDataBlock; ()	Obtain the tlv data of NFC

setCardTradeMode

Item	Details
Signature	-(void)setCardTradeMode:(CardTradeMode) aCardTMode;
Inputs	CardTradeMode_ONLY_INSERT_CARD, CardTradeMode_ONLY_SWIPE_CARD, CardTradeMode_SWIPE_INSERT_CARD, CardTradeMode_UNALLOWED_LOW_TRADE, CardTradeMode_SWIPE_TAP_INSERT_CARD,// add 20150715 CardTradeMode_SWIPE_TAP_INSERT_CARD_UNALLOWED_LOW_TRADE, CardTradeMode_ONLY_TAP_CARD, CardTradeMode_SWIPE_TAP_INSERT_CARD_NOTUP, CardTradeMode_TAP_INSERT_CARD_NOTUP, CardTradeMode_TAP_INSERT_CARD_TUP, CardTradeMode_SWIPE_TAP_INSERT_CARD_Down, CardTradeMode_SWIPE_TAP_INSERT_CARD_NOTUP_UNALLOWED_LOW_TRADE, CardTradeMode_SWIPE_INSERT_CARD_UNALLOWED_LOW_TRADE, CardTradeMode_SWIPE_TAP_INSERT_CARD_UNALLOWED_LOW_TRADE_NEW, CardTradeMode_SWIPE_TAP_INSERT_CARD_NOTUP_DELAY,
Outputs	None
Example	[pos setCardTradeMode:CardTradeMode_SWIPE_TAP_INSERT_CARD_NOTUP]

doTrade

Item	Details
Signature	-(void) doTrade:(NSInteger) timeout;
Inputs	Timeout
Outputs	None
Callback	onDoTradeResult
Example	[pos doTrade:60]

setAmountPoint

Item	Details
Signature	-(void)setAmountPoint:(BOOL)amoutPoint;
Inputs	Amout point
Outputs	None
Example	[pos setAmountPoint:true]

cancelSetAmount

Item	Details
Signature	-(void)cancelSetAmount()
Inputs	None
Outputs	None
Example	[pos cancelSetAmount]

sendTime

Item	Details
Signature	-(void)sendTime: (NSString *)aterminalTime;
Inputs	Terminal time

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Item	Details
Outputs	None
Example	[pos sendTime:terminalTime]

doEmvApp

Item	Details
Signature	-(void) doEmvApp: (EmvOption)aemvOption;
Inputs	EmvOption_START, EmvOption_START_WITH_FORCE_ONLINE, EmvOption_START_WITH_FORCE_PIN, EmvOption_START_WITH_FORCE_ONLINE_FORCE_PIN
Outputs	None
Callback	onDoTradeResult
Example	[pos doEmvApp:EmvOption.START]

selectEmvApp

Item	Details
Signature	-(void) selectEmvApp: (NSInteger)index;
Inputs	Index: select emv app index
Outputs	None
Callback	onRequestSelectEmvApp
Example	[pos selectEmvApp:index]

cancelSelectEmvApp

Item	Details
Signature	void cancelSelectEmvApp()
Inputs	None

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Item	Details
Outputs	None
Callback	onRequestSelectEmvApp
Example	[pos cancelSelectEmvApp]

sendPinEntryResult

Item	Details
Signature	-(void)sendPinEntryResult:(NSString *)pin;
Inputs	pin: pin code length just allows 4-12
Outputs	None
Example	[pos sendPinEntryResult:pin]

cancelPinEntry

Item	Details
Signature	-(void)cancelPinEntry;
Inputs	None
Outputs	None
Example	[pos cancelPinEntry]

sendOnlineProcessResult

Item	Details
Signature	-(void) sendOnlineProcessResult: (NSString *)tlv;
Inputs	tlv data consist of 8A+91+71/72 8A: Authorisation Response Code 91: Issuer Authentication Data 71/72: Issuer Script
Outputs	None

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Item	Details
Callback	onRequestOnlineProcess
Example	[pos sendOnlineProcessResult:@" 8A023030910A6C741A0100F69FF90012720F860D842400 00082129027736EDDA04"];

getNFCBatchData

Item	Details
Signature	-(void)getNFCBatchData:(void(^)(NSDictionary *dict))nfcBatchDataBlock;
Inputs	None
Outputs	Return the TLV data of NFC
Callback	None
Example	[pos getNFCBatchData:^(NSDictionary *dict) { NSString *tlvData = dict[@"tlv"] }];

4.4.3 EMV Configuration APIs

The following describes the emv config update interface in detail.

API Methods	Description
(void)updateEMVConfigByXml:(NSString *)xmlStr;	Update the emv config

updateEMVConfigByXml

Item	Details
Signature	-(void)updateEMVConfigByXml:(NSString *)xmlStr;
Inputs	xmlStr: parse EMV configuration file in xml format to get string
Outputs	None
Callback	onReturnCustomConfigResult

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Item	Details
Example	<pre> NSData *xmlData = [self readLine:@"emv_profile_tlv"]; NSString *xmlStr =[QPOSUtil asciiFormatString:xmlData]; [pos updateEMVConfigByXml:xmlStr]; </pre>

4.4.4 Key Management APIs

These APIs are used to update encryption keys (e.g., Master Key, Work Key, IPEK) on the QPOS device, ensuring transaction security.

API Methods	Description
(void)updateKeyByTR_31:(NSInteger)keyIndex keyBlock:(NSString *)keyBlock;	Update the device key according analysis the TR31 keyblock

updateKeyByTR_31

Item	Details
Signature	-(void)updateKeyByTR_31:(NSInteger)keyIndex keyBlock:(NSString *)keyBlock;
Inputs	keyIndex: new key index, from 0 start keyBlock: TR31 keybolck
Outputs	None
Callback	onReturnUpdateKeyByTR_31Result
Example	<pre> [pos updateKeyByTR_31:0 keyBlock:@"B0080B1TX00E00007ADC3F4EC9DEC96726CAB25032AC86B715814 29795F43E2B6B5DBCD646FE8F3A"] </pre>

4.4.5 Firmware Update APIs

These APIs are used to update the firmware of the QPOS device, ensuring compatibility with new features and security patches.

updatePosFirmware

Item	Details
Signature	-(NSInteger)updatePosFirmware:(NSData*)aData address:(NSString*)devAddress;
Inputs	data: the firmware asc file data. Device Address: device address, like Bluetooth connection, need input the device Bluetooth mac address, other connection method, can input "".
Outputs	Return int value, -1 means error, 0 means can update.
Description	Update the device firmware
Callback	onUpdatePosFirmwareResult

getUpdateProgress

Item	Details
Signature	-(NSInteger)getUpdateProgress;
Inputs	None
Outputs	Return the firmware update progress integer value
Description	Get the device firmware update progress. And it should be called in the child thread so that avoid to block the main thread.

4.4.6 Device Information APIs

These APIs are used to retrieve hardware and software information from the QPOS device.

getQposId

Item	Details
Signature	void getQposId()
Inputs	None
Outputs	None
Description	Retrieve the POS_ID of the QPOS device. Results are returned by onQposIdResult.

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Item	Details
Callback	onQposIdResult

getQposInfo

Item	Details
Signature	void getQposInfo()
Inputs	None
Outputs	None
Description	Get the device info
Callback	onQposInfoResult

4.5 Delegate Methods Reference

Delegate methods (callbacks) are triggered by the SDK to notify the application of event status (e.g., connection results, key update results). All callbacks are implemented via the 'QPOSServiceListener' interface.

4.5.1 Connection Callbacks

These callbacks notify the application of device connection/disconnection status.

onRequestQposConnected

Item	Details
Signature	-(void)onRequestQposConnected ()
Inputs	None
Outputs	None
Description	Callback of connected success

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onRequestQposDisconnected

Item	Details
Signature	-(void)onRequestQposDisconnected()
Inputs	None
Outputs	None
Description	Callback of disconnect

onRequestNoQposDetected

Item	Details
Signature	-(void)onRequestNoQposDetected()
Inputs	None
Outputs	None
Description	Callback of connected fail

onBluetoothName2Mode

Item	Details
Signature	-(void)onBluetoothName2Mode:(NSString *)bluetoothName
Inputs	Return the scanned Bluetooth names
Outputs	None
Description	Return the scanned Bluetooth names

4.5.2 Transaction Callbacks

onRequestSetAmount

Item	Details
Signature	void onRequestSetAmount()

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Item	Details
Inputs	None
Outputs	None
Description	Callback of setAmount

onRequestTime

Item	Details
Signature	-(void)onRequestTime()
Inputs	None
Outputs	None
Description	Callback method of request user to set time

onRequestSelectEmvApp

Item	Details
Signature	-(void)onRequestSelectEmvApp: (NSArray*)appList;
Inputs	appList
Outputs	None
Description	Callback method of request user to select emv app

onDoTradeResult

Item	Details
Signature	-(void)onDoTradeResult: (DoTradeResult)result DecodeData:(NSDictionary*)decodeData;

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Item	Details
Inputs	result: DoTradeResult.NONE, MCR, ICC, NOT_ICC, BAD_SWIPE, NO_RESPONSE, NO_UPDATE_WORK_KEY, NFC_ONLINE, NFC_OFFLINE, NFC_DECLINED, TRY_ANOTHER_INTERFACE, CARD_NOT_SUPPORT, PLS_SEE_PHONE, REQUEST_ONLINE decodeData: the transaction data
Outputs	None
Description	Returns transaction data for magnetic stripe and NFC cards, and poll card results for ICC cards.

onRequestOnlineProcess

Item	Details
Signature	-(void)onRequestOnlineProcess: (NSString*) tlv;
Inputs	tlv: contains the tag-length-value data structure returned by the EMV kernel
Outputs	None
Description	EMV kernel request the online handler, should send the results back to EMV kernel by sendOnlineProcessResult()

onRequestTransactionResult

Item	Details
Signature	-(void)onRequestTransactionResult: (TransactionResult)transactionResult;

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Item	Details
Inputs	transactionResult: TransactionResult. APPROVED, TERMINATED, DECLINED, CANCEL, CAPK_FAIL, NOT_ICC, SELECT_APP_FAIL, DEVICE_ERROR, CARD_NOT_SUPPORTED, MISSING_MANDATORY_DATA, CARD_BLOCKED_OR_NO_EMV_APPS, INVALID_ICC_DATA, FALLBACK, NFC_TERMINATED, CARD_REMOVED, TRADE_LOG_FULL, TRANSACTION_NOT_ALLOWED_AMOUNT_EXCEED, CONTACTLESS_TRANSACTION_NOT_ALLOW, TRANS_TOKEN_INVALID, CARD_BLOCKED, APP_BLOCKED, MULTIPLE_CARDS
Outputs	None
Description	Transaction result response

onRequestBatchData

Item	Details
Signature	-(void)onRequestBatchData: (NSString*)tlv;
Inputs	tlv: contains the tag-length-value data structure returned by the EMV kernel
Outputs	None
Description	Response of ICC transactions: when the transaction is finished. The batch data will be returned to the application

onReturnReversalData

Item	Details
Signature	-(void)onReturnReversalData: (NSString*)tlv;
Inputs	tlv : contains the tag-length-value data structure returned by the EMV kernel
Outputs	None
Description	IC card reversal data response

onEmvICCExceptionData

Item	Details
Signature	-(void)onEmvICCExceptionData: (NSString*)tlv;
Inputs	tlv : contains the tag-length-value data structure returned by the EMV kernel
Outputs	None
Description	EMV ICC Exception response

4.5.3 EMV Configuration Callbacks**onReturnCustomConfigResult**

Item	Details
Signature	-(void)onReturnCustomConfigResult:(BOOL)isSuccess config:(NSString*)resultl;
Inputs	isSuccess : result of the emv update result : if update failed, it will return the specific error reasons
Outputs	None
Description	Callback of updateEMVConfigByXml()

4.5.4 Key Management Callbacks

These callbacks notify the application of key update status.

onReturnUpdateKeyByTR_31Result

Item	Details
Signature	-(void)onReturnUpdateKeyByTR_31Result:(BOOL)result;
Inputs	result: true– update key success, false – update key failure
Outputs	None
Description	Callback of updateKeyByTR_31()

4.5.5 Firmware Update Callbacks**onUpdatePosFirmwareResult**

Item	Details
Signature	-(void)onUpdatePosFirmwareResult:(UpdateInformationResult)result;
Inputs	result: UPDATE_SUCCESS UPDATE_FAIL UPDATE_PACKET_VEFIRY_ERROR //return Update firmware package is wrong
Outputs	None
Description	Callback of updatePosFirmware()

4.5.6 Device Information Callbacks

These callbacks return the device information retrieved by the SDK.

onQposIdResult

Item	Details
Signature	-(void)onQposIdResult: (NSDictionary*)posId;

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Item	Details
Inputs	posId:27000102925052100100 csn: conn: true psamId: 3237303630303630 NFCId: FF00
Outputs	None
Description	Callback of getQposId()

onQposInfoResult

Item	Details
Signature	-(void)onQposInfoResult: (NSDictionary*)posInfoData;
Inputs	Bootloader Version:3.1 Firmware Version:6.27.4.15 USB: false Charge: false Battery ADC:4044 mV battery percentage:100% Hardware Version: A27C P1 SUB : Track 1 Supported:true Track 2 Supported:true Track 3 Supported:false PCI FirmwareVresion:1.4.0 PCI HardwareVersion:2.3.1 compileTime:20251126102744
Outputs	None
Description	Callback of getQposInfo()